29 Mixing in reactors and residence time distributions

Why Are We Learning This?

So far in reactor design, we have analyzed **ideal reactors**. We assumed that reactors behave exactly like the mathematical models used in our mole balances:

- Batch reactors are perfectly mixed.
- CSTRs are perfectly mixed.
- PFRs are perfectly mixed in the radial dimension and have no axial mixing.
- PBRs behave like ideal plug-flow systems.

These assumptions make reactor design possible, but real reactors rarely behave perfectly. In practice:

- Some molecules spend longer in a reactor than others.
- Mixing is often incomplete.
- Flow patterns deviate from ideal behavior.
- Dead zones, recirculation zones, and bypassing may occur.

As a result, real reactor performance is often different from the predictions made using ideal-reactor models. By the end of this lecture, you should be able to:

- Explain why real reactors deviate from ideal behavior.
- Understand the concept of residence time.
- Define a residence time distribution (RTD).
- Explain why RTDs arise in real reactors.
- Compare RTDs for ideal and non-ideal reactors.
- Understand how imperfect mixing affects reactor conversion.

Course Roadmap

Introduction to CRE

↓

General Balance Equation

↓

Common Reactor Models
(CSTR, Batch, PFR)

↓

Reaction Progress
(Conversion)

↓

Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)

↓

Rate Laws
(Kinetics)

↓

Equilibrium
(Reversible Reactions)

↓

Equilibrium Conversion
(Thermodynamic Limitations)

↓

One Reaction: Solving in terms of Reaction Progress

↓

One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)

↓

Catalytic Reactions

↓

Numerical Methods
(Python)

↓

Energy Balances

↓

One Reaction: Non-Isothermal Reactor Design

↓

Multiple Reactions

↓

Reactor Safety

↓

►Challenging the Well-Mixed Assumption◄
(Residence Time Distributions)

↓

Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Big Idea

Ideal Reactor
↓
One Residence Time
↓
Predictable Conversion

Real Reactor
↓
Many Residence Times
↓
Altered Conversion

This lecture begins the transition from ideal reactor models toward understanding how real reactors actually behave.

So far in this class, we have studied ideal reactors, i.e., where we have assumed

For example:

- In the batch reactor and CSTR, we have assumed

- In the PFR and PBR, we have assumed

What Does "Ideal Reactor" Really Mean?

An ideal reactor is not necessarily a realistic reactor. Instead, an ideal reactor is a mathematical model that makes specific assumptions about mixing and flow.

Examples:

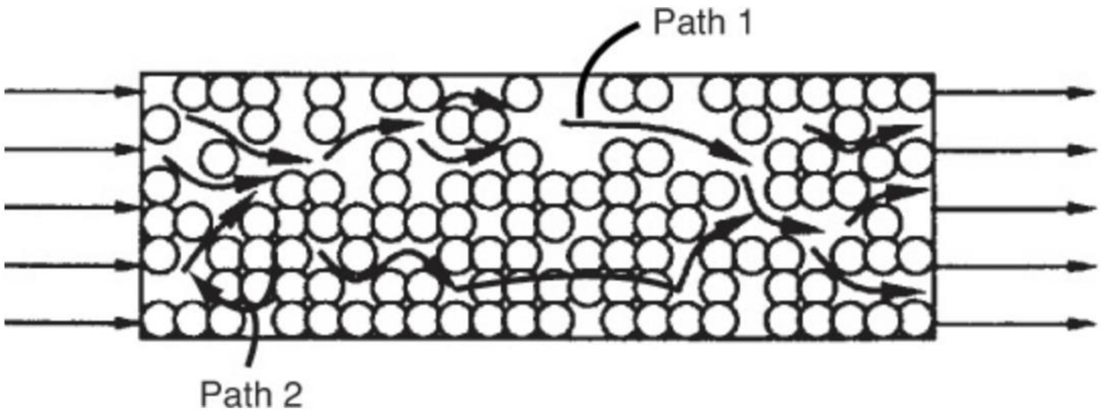
Ideal Batch Reactor: Perfect Mixing, No Inlet, No Outlet

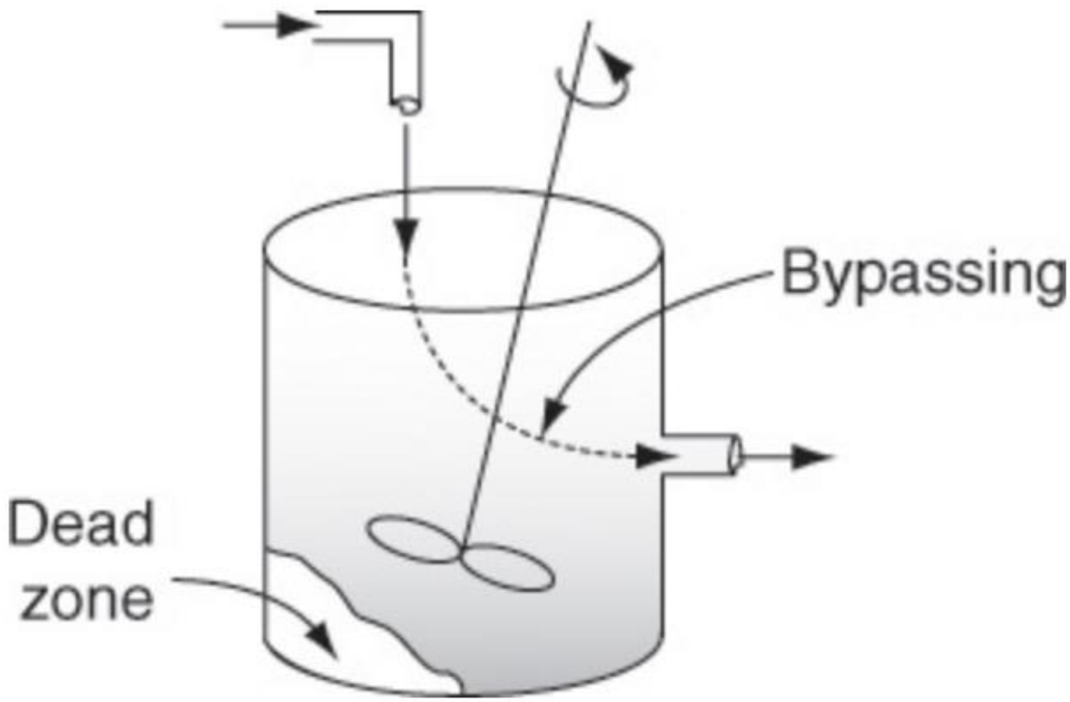
Ideal CSTR: Perfect Mixing, Uniform Composition Throughout the Reactor

Ideal PFR: Perfect Radial Mixing, No Axial Mixing, Fluid Elements Move Like a Plug

These ideal models allow reactor design equations to be derived and solved.

Real reactors are generally not perfectly mixed. For example:





[credit for images]

Why Mixing Matters

Imagine two molecules entering the same reactor. In a real reactor:

Molecule 1 may leave quickly
Molecule 2 may remain much longer

Even though both molecules entered at the same time, they experience different reaction times.

Physical Interpretation

Different Residence Times
↓
Different Conversions
↓
Different Reactor Performance

This is the fundamental reason we study residence time distributions.

In these examples, different molecules spend amounts of time in the reactor

In ideal PFRs, PBRs, and batch reactors, all the molecules spend amount of time in the reactor

- Batch:

- PFR/PBR:

In contrast, there is a distribution of residence times in a CSTR:

Gives rise to the concept of residence time distribution (RTD):

Residence time distributions also occur in non-ideal PFRs/PBRs.

What Is a Residence Time Distribution?

Residence time refers to: How Long A Molecule Spends Inside a Reactor

In ideal reactors: Residence Time is Predictable.

In real reactors: Some Molecules Leave Early, Some Leave Late.

This creates a distribution of residence times called the: Residence Time Distribution (RTD)

Why RTDs Are Important

RTDs help answer questions such as: Is the reactor well mixed? Is there bypassing? Are dead zones present? Does the reactor behave ideally?

RTDs provide a quantitative measure of reactor mixing behavior.

Physical Interpretation

Different Molecules Have Different Histories Inside the Reactor. Some molecules may: Leave Too Early while others may Remain Much Longer. Both situations affect reactor performance and conversion.

Engineering Significance

An RTD allows us to compare: Real Reactor Behavior with Ideal Reactor Behavior and determine whether the reactor suffers from: Poor Mixing, Bypassing, Dead Zones, Recirculation.

Understanding the RTD is often the first step toward diagnosing and improving reactor performance.

Common Non-Ideal Flow Problems

Several flow problems can create non-ideal behavior:

Bypassing: Fluid Leaves Too Quickly. Result: Lower Conversion

Dead Zones: Fluid Becomes Trapped. Result: Poor Utilization of Reactor Volume

Recirculation: Fluid Moves in Loops. Result: Broad Residence Time Distribution

These effects cause real reactor performance to differ from ideal-reactor predictions.

Conversions in real reactors are usually than in ideal reactors.

Lower conversions are caused by ineffective

Why Non-Ideal Mixing Reduces Conversion

Ideal reactor models assume that every reactor volume element is being used effectively. In real reactors:

Some Regions May Be Underutilized

Some Molecules May Leave Too Soon

Some Regions May Be Poorly Mixed

As a result: Actual Conversion < Ideal Conversion

in many practical situations.

Engineering Interpretation

Poor Mixing
↓
Less Effective Contact Time
↓
Less Reaction
↓
Lower Conversion

This is one of the primary motivations for studying RTDs.

Summary and Key Takeaways

Today we began studying how real reactors differ from ideal reactor models.

Key Ideas

- ☑ Ideal reactors assume specific mixing and flow behavior.
- ☑ Real reactors are generally not perfectly mixed.
- ☑ Different molecules may spend different amounts of time inside a reactor.
- ☑ This variation creates a residence time distribution (RTD).
- ☑ RTDs occur in both CSTRs and non-ideal plug-flow systems.

☑ Poor mixing often reduces conversion relative to ideal reactor predictions.

☑ RTDs are one of the primary tools used to characterize reactor mixing behavior.

Most Important Idea

Ideal Reactor
↓
Known Residence Time
↓
Predictable Performance

Real Reactor
↓
Residence Time Distribution
↓
Non-Ideal Performance

The central lesson of this lecture is that reactor performance depends not only on kinetics and stoichiometry, but also on how fluid actually moves and mixes inside the reactor.